SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

TUTORIAL ASSIGNMENT - 2
DATA WAREHOUSING AND DATA MINING

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Batch : III Year / V Sem / BDA "C"

Subject : Data Warehousing and Data Mining

Assignment Description

This tutorial assignment was performed using Orange Data Mining, a visual data mining and machine learning tool. The practicals demonstrate data preparation, transformation, association rule mining, statistical analysis, regression, outlier analysis, and clustering through Orange widgets and visual workflows.

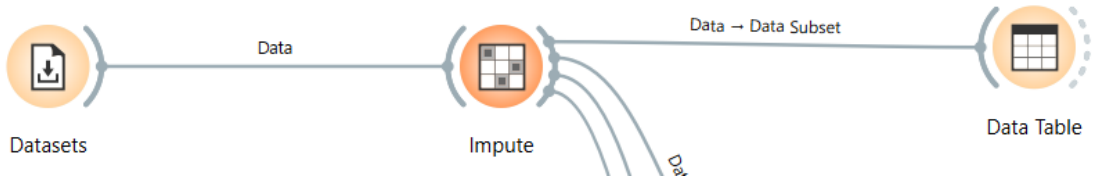
Experiments Included

1. Data Cleaning on Dataset using Orange
2. Data Reduction on Dataset using Orange
3. Data Transformation using Orange
4. Generate Association Rules using Apriori Algorithm
5. Statistical Analysis on Dataset using Orange
6. Performing Regression on Dataset
7. Implementation of Outlier Analysis using Orange
8. K-Means Algorithm for Iris Dataset
9. Clustering on iris.arff using Simple K-Means

Submitted as part of the Data Warehousing and Data Mining Tutorial Assignment - 2.

1: Data cleaning using Orange

Aim: To perform data cleaning on a dataset using Orange Data Mining.



Steps

Output:

	disease state	--Control	1949_at	31397_at	31418_at	31420_at	31423_at	31430_at	31456_at	31588_at	31655_at	31709_at	31710_at
1	docetaxel resist...	850.384	12.003	38.887	138.804	24.600	313.709	25.422	5.898	95.546	48.030	120.784	256
2	docetaxel resist...	606.788	8.435	52.092	102.492	15.783	180.236	22.545	5.898	75.791	244.456	147.660	155
3	docetaxel resist...	537.786	17.764	52.553	128.517	20.674	205.834	11.911	5.898	64.893	261.610	161.733	155
4	docetaxel resist...	553.825	17.306	30.746	162.779	33.528	149.981	24.833	5.898	115.181	25.593	166.724	299
5	docetaxel resist...	567.721	24.683	50.110	96.141	19.285	93.800	19.150	6.521	64.894	627.125	153.084	136
6	docetaxel resist...	706.714	12.765	19.149	148.980	23.315	246.499	23.600	5.898	118.900	61.429	291.231	299
7	docetaxel resist...	856.824	11.647	26.802	132.972	32.803	318.081	5.898	21.691	77.137	309.116	188.034	190
8	docetaxel resist...	554.025	11.982	39.472	121.922	21.213	169.507	22.151	5.898	72.808	911.636	188.735	199
9	docetaxel resist...	403.512	7.973	39.368	137.465	13.874	167.567	11.032	12.160	42.126	735.625	147.779	160
10	docetaxel resist...	547.932	6.349	24.588	155.837	19.860	137.418	14.748	5.898	64.293	429.854	210.593	163
11	docetaxel resist...	512.349	5.898	30.014	95.731	16.490	185.316	10.365	14.812	69.258	597.966	165.156	142
12	docetaxel resist...	754.697	27.434	11.248	112.399	39.145	448.683	19.849	8.231	77.010	374.496	316.037	296
13	docetaxel resist...	751.797	16.614	24.086	104.646	38.402	434.608	11.156	15.581	97.587	711.474	269.137	268
14	docetaxel resist...	602.149	5.898	15.750	171.559	20.296	190.520	9.681	23.031	94.402	365.641	334.092	197
15	docetaxel sensit...	507.783	9.691	34.650	99.463	5.898	72.522	11.597	5.898	76.459	753.375	160.558	122
16	docetaxel sensit...	434.991	17.590	56.640	98.195	14.613	79.093	16.617	5.898	88.805	565.160	127.289	127
17	docetaxel sensit...	543.119	10.261	32.490	102.310	12.013	105.533	13.564	13.142	61.668	380.579	174.935	156
18	docetaxel sensit...	523.310	11.953	25.252	126.307	14.685	332.197	19.586	13.359	80.709	690.752	125.090	135
19	docetaxel sensit...	441.479	9.826	37.612	136.070	9.627	112.414	5.898	10.374	49.581	447.989	142.801	136
20	docetaxel sensit...	555.931	6.686	16.583	100.592	24.067	196.904	10.380	9.383	94.356	230.441	140.561	174
21	docetaxel sensit...	481.638	5.898	18.388	106.423	5.898	202.605	14.774	5.898	47.443	605.373	156.775	115
22	docetaxel sensit...	853.811	5.898	17.120	190.768	6.488	202.763	5.898	5.898	80.688	342.491	153.069	198
23	docetaxel sensit...	995.558	11.636	19.494	120.793	10.174	137.405	20.787	13.522	73.449	642.073	217.001	176
24	docetaxel sensit...	476.480	6.763	30.882	223.352	11.602	405.067	10.828	20.038	59.023	458.538	148.513	168

Result: Data Cleaning using Python was achieved

2: Data Reduction on Dataset using Orange

Aim: To perform data reduction on a dataset using Orange Data Mining by removing an unnecessary attribute.



Steps

Output:

	category	content	Title	Speaker_party	peaker_party_nam	Speaker_gender
• 1	2020	I can reassure ...	Minutes of the ...	CON	Conservative	M
• 2	2020	I thank both th...	Minutes of the ...	LAB	Labour	M
• 3	2019	Further to that ...	Minutes of the ...	LAB	Labour	M
• 4	2019	The hon. Lady ...	Minutes of the ...	CON	Conservative	M
• 5	2019	Amen to that, ...	Minutes of the ...	CON	Conservative	M
• 6	2019	The noble Baro...	Minutes of the ...	LAB	Labour	M
• 7	2020	I am sure that t...	Minutes of the ...	PC	Plaid Cymru	F
• 8	2019	My Lords, I beg...	Minutes of the ...	LD	Liberal Democrat	M
• 9	2019	My hon. Friend ...	Minutes of the ...	CON	Conservative	M
• 10	2019	Does the hon. L...	Minutes of the ...	CON	Conservative	M
• 11	2019	I beg to move, ...	Minutes of the ...	LAB	Labour	F
• 12	2020	My hon. Friend ...	Minutes of the ...	CON	Conservative	M
• 13	2020	Next year will s...	Minutes of the ...	LD	Liberal Democrat	F
• 14	2019	I thank the Ho...	Minutes of the ...	SNP	Scottish Nation...	F
• 15	2020	My Lords, in th...	Minutes of the ...	CON	Conservative	F
• 16	2020	The goal of the ...	Minutes of the ...	CON	Conservative	M
• 17	2019	I thank the Spe...	Minutes of the ...	CON;I	Conservative;In...	F
• 18	2020	I thank the hon....	Minutes of the ...	CON	Conservative	F
• 19	2019	My Lords, the T...	Minutes of the ...	LD	Liberal Democrat	M
• 20	2020	My Lords, over ...	Minutes of the ...	LD	Liberal Democrat	F
• 21	2019	Child poverty h...	Minutes of the ...	LAB	Labour	M
• 22	2020	The noble Lord ...	Minutes of the ...	CON	Conservative	M
• 23	2019	My Lords, like ...	Minutes of the ...	LAB	Labour	M
• 24	2020	My Lords, I shar...	Minutes of the ...	CB	Crossbench	F
• 25	2019	Those who clai...	Minutes of the ...	LAB	Labour	F
• 26	2020	The hon. Gentle...	Minutes of the ...	CON	Conservative	F
• 27	2020	First, when it co...	Minutes of the ...	CON	Conservative	F
• 28	2020	I will call Memb...	Minutes of the ...	LAB	Labour	M
• 29	2019	Further to that ...	Minutes of the ...	IGC	The Independe...	M
• 30	2019	I rise to speak b...	Minutes of the ...	CON	Conservative	F
• 31	2020	My hon. Friend ...	Minutes of the ...	CON	Conservative	F
• 32	2020	I pay warm trib...	Minutes of the ...	CON	Conservative	M
• 33	2019	My understandi...	Minutes of the ...	CON	Conservative	F
• 34	2019	Further to the ...	Minutes of the ...	CON	Conservative	F
• 35	2020	My Lords, I agr...	Minutes of the ...	LD;NA	Liberal Democr...	F
• 36	2019	My hon. Friend ...	Minutes of the ...	CON	Conservative	M

Result: Data Reduction on Dataset using Orange was achieved.

3: Data Transformation Using Orange

Aim : To perform data transformation and normalization using Orange Data Mining by selecting relevant columns and converting numerical attributes into a normalized range.



Steps

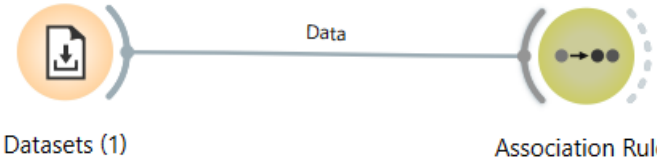
Output:

	body fat brozek	id	age	weight	adiposity	neck	abdomen	tight	knee
1	12.6	1	-1.740	-0.84125	-0.4771	-0.739	-0.6835	-0.0775	-0.536
2	6.9	2	-1.820	-0.19346	-0.5595	0.209	-0.8880	-0.1347	-0.536
3	24.6	3	-1.820	-0.84977	-0.2024	-1.645	-0.4326	0.0370	0.129
4	10.9	4	-1.502	0.19862	-0.1475	-0.244	-0.5720	0.1325	-0.536
5	27.8	5	-1.661	0.18157	0.0448	-1.481	0.6917	0.7241	1.500
6	20.6	6	-1.661	1.06801	0.2920	0.415	0.1714	1.2585	1.416
7	19.0	7	-1.502	0.07077	0.2096	-0.656	-0.1725	-0.1920	-0.121
8	12.8	8	-1.581	-0.09970	-0.5045	-0.079	-0.3769	0.1134	0.336
9	5.1	9	-1.581	0.41170	-0.2299	0.044	-0.9344	0.6669	-0.121
10	12.0	10	-1.740	0.65888	0.0997	1.693	-0.3676	0.7050	1.292
11	7.5	11	-1.502	0.24976	-0.5045	0.209	-0.8322	0.0561	0.461
12	8.5	12	-1.422	1.26405	0.2371	0.580	-0.1539	1.2967	0.253
13	20.5	13	-1.024	0.05372	0.2371	0.168	-0.0888	0.7623	-0.121
14	20.8	14	-1.184	0.89754	0.8413	0.580	0.8590	1.2585	1.209
15	21.7	15	-0.786	0.30090	0.5392	1.034	0.3572	1.8311	0.170
16	20.5	16	-0.786	-0.55145	0.2371	-0.656	0.0227	0.7050	0.046
17	28.1	17	-0.865	0.57365	0.5117	0.374	0.3572	1.0295	0.918
18	22.4	18	-1.024	1.03392	1.0336	1.693	0.4594	1.4303	0.586
19	16.1	19	-1.343	0.16452	0.7589	0.003	-0.2747	0.9150	0.046
20	16.5	20	-0.945	1.11915	0.5941	0.828	0.7382	1.2203	0.835
21	19.0	21	-1.343	0.00258	0.5117	0.457	0.3107	0.7814	-0.245
22	15.3	22	-1.343	0.73560	1.0061	1.363	0.5802	0.7623	0.835
23	15.7	23	-1.104	-1.31856	-1.1637	-1.687	-1.5013	-0.3829	-1.367
24	17.6	24	-1.024	-1.02876	-1.1088	-1.027	-1.1667	-0.8600	-0.993
25	14.2	25	-1.343	-0.94353	-0.6144	-1.439	-1.5105	-0.1920	-1.284
26	4.6	26	-1.422	-0.67078	-0.9714	-0.945	-1.1946	-0.8409	-0.785
27	8.5	27	-0.865	-1.61688	-1.4109	-0.739	-1.6685	-1.4707	-1.616
28	22.4	28	-1.104	-1.05433	-0.6968	0.333	-0.3583	-0.3638	-1.076
29	4.7	29	-1.422	-1.55722	-0.8341	-0.656	-1.7336	-1.7761	-1.699
30	9.4	30	-1.263	-0.61964	-0.4496	-0.533	-0.8415	-0.0966	-1.367
31	12.3	31	-1.024	0.10486	-0.5045	0.292	-0.3583	-0.3638	0.046
32	6.5	32	-1.263	-0.63668	-0.8890	-0.285	-0.7486	-0.1729	0.087
33	13.4	33	-1.422	-0.37246	-0.5869	0.044	-1.2504	-0.4019	-0.993
34	20.9	34	-0.309	1.34929	1.3906	0.745	0.7382	1.4685	2.330
35	31.1	35	-0.309	2.32948	1.8576	1.693	2.1413	2.2510	1.957
36	38.2	36	0.327	0.43727	1.8026	0.168	1.9090	0.4760	-0.121

Result: Data Transformation Using Orange

4: Generate Association Rules using Apriori Algorithm

Aim: To generate association rules from a transactional dataset using the Apriori algorithm in Orange Data Mining.



Steps

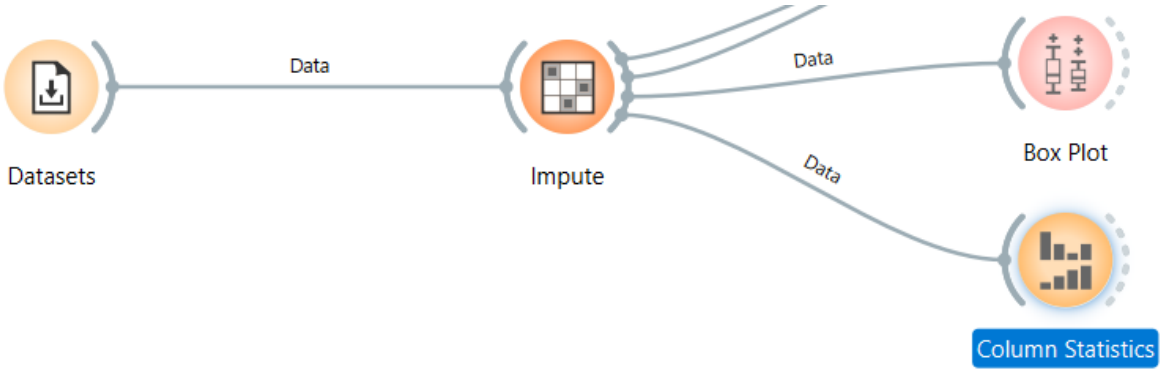
Output:

Supp	Conf	Covr	Strg	Lift	Levr	Antecedent		
0.600	1.000	0.600	1.333	1.250	0.120	Beer=1	→	Diapers=1
0.400	1.000	0.400	2.000	1.250	0.080	Bread=1, Beer=1	→	Diapers=1
0.400	1.000	0.400	2.000	1.250	0.080	Milk=1, Beer=1	→	Diapers=1
0.200	1.000	0.200	4.000	1.250	0.040	Bread=1, Milk=1, Beer=1	→	Diapers=1
0.200	1.000	0.200	4.000	1.250	0.040	Eggs=1	→	Bread=1
0.200	1.000	0.200	4.000	1.250	0.040	Eggs=1	→	Diapers=1
0.200	1.000	0.200	4.000	1.250	0.040	Diapers=1, Eggs=1	→	Bread=1
0.200	1.000	0.200	4.000	1.250	0.040	Bread=1, Eggs=1	→	Diapers=1
0.200	1.000	0.200	3.000	1.667	0.080	Eggs=1	→	Bread=1, Diapers=1
0.200	1.000	0.200	3.000	1.667	0.080	Eggs=1	→	Beer=1
0.200	1.000	0.200	4.000	1.250	0.040	Beer=1, Eggs=1	→	Bread=1
0.200	1.000	0.200	3.000	1.667	0.080	Bread=1, Eggs=1	→	Beer=1
0.200	1.000	0.200	2.000	2.500	0.120	Eggs=1	→	Bread=1, Beer=1
0.200	1.000	0.200	4.000	1.250	0.040	Beer=1, Eggs=1	→	Diapers=1
0.200	1.000	0.200	3.000	1.667	0.080	Diapers=1, Eggs=1	→	Beer=1
0.200	1.000	0.200	3.000	1.667	0.080	Eggs=1	→	Diapers=1, Beer=1
0.200	1.000	0.200	4.000	1.250	0.040	Diapers=1, Beer=1, Eggs=1	→	Bread=1
0.200	1.000	0.200	4.000	1.250	0.040	Bread=1, Beer=1, Eggs=1	→	Diapers=1
0.200	1.000	0.200	3.000	1.667	0.080	Beer=1, Eggs=1	→	Bread=1, Diapers=1
0.200	1.000	0.200	3.000	1.667	0.080	Bread=1, Diapers=1, Eggs=1	→	Beer=1
0.200	1.000	0.200	2.000	2.500	0.120	Diapers=1, Eggs=1	→	Bread=1, Beer=1
0.200	1.000	0.200	3.000	1.667	0.080	Bread=1, Eggs=1	→	Diapers=1, Beer=1
0.200	1.000	0.200	2.000	2.500	0.120	Eggs=1	→	Bread=1, Diapers=1, Beer=1
0.400	1.000	0.400	2.000	1.250	0.080	Cola=1	→	Milk=1
0.200	1.000	0.200	4.000	1.250	0.040	Bread=1, Cola=1	→	Milk=1
0.400	1.000	0.400	2.000	1.250	0.080	Cola=1	→	Diapers=1
0.200	1.000	0.200	4.000	1.250	0.040	Bread=1, Cola=1	→	Diapers=1
0.400	1.000	0.400	2.000	1.250	0.080	Diapers=1, Cola=1	→	Milk=1
0.400	1.000	0.400	2.000	1.250	0.080	Milk=1, Cola=1	→	Diapers=1
0.400	1.000	0.400	1.500	1.667	0.160	Cola=1	→	Milk=1, Diapers=1

Result: Association Rules were generated using the Apriori Algorithm in Orange.

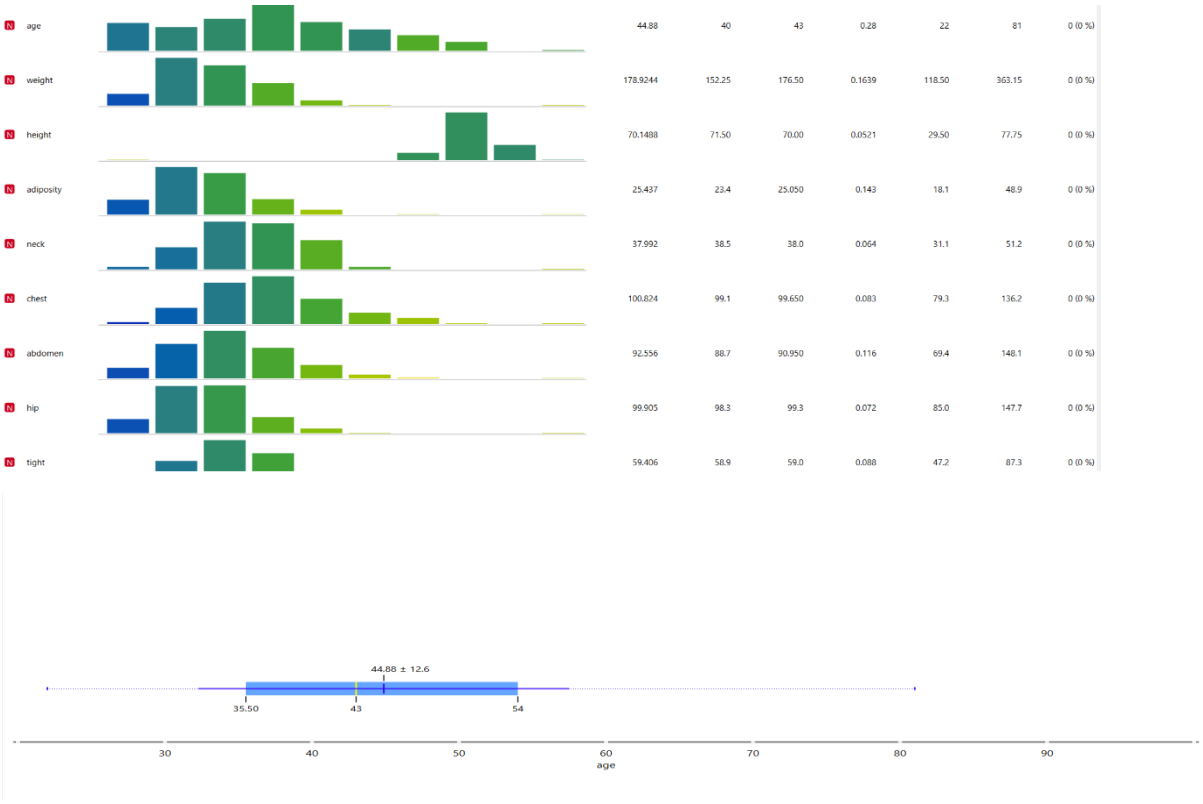
5: Statistical Analysis on Dataset using Orange

Aim: To perform statistical analysis on a dataset using Orange Data Mining by computing column statistics and visualizing the data distribution with a box plot.



Steps

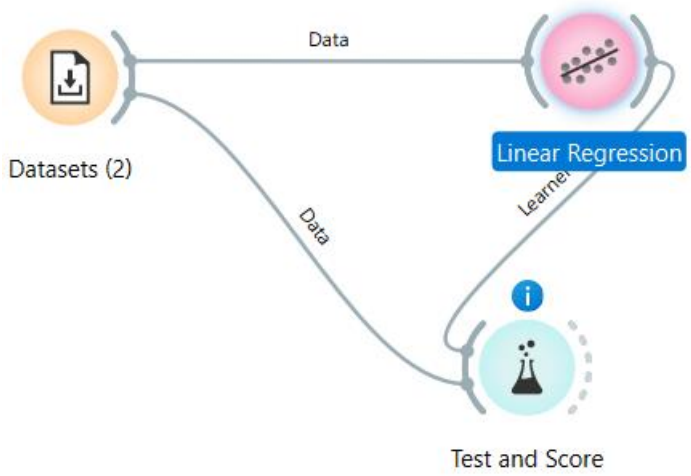
Output:



Result: Statistical Analysis on Dataset using Orange was achieved.

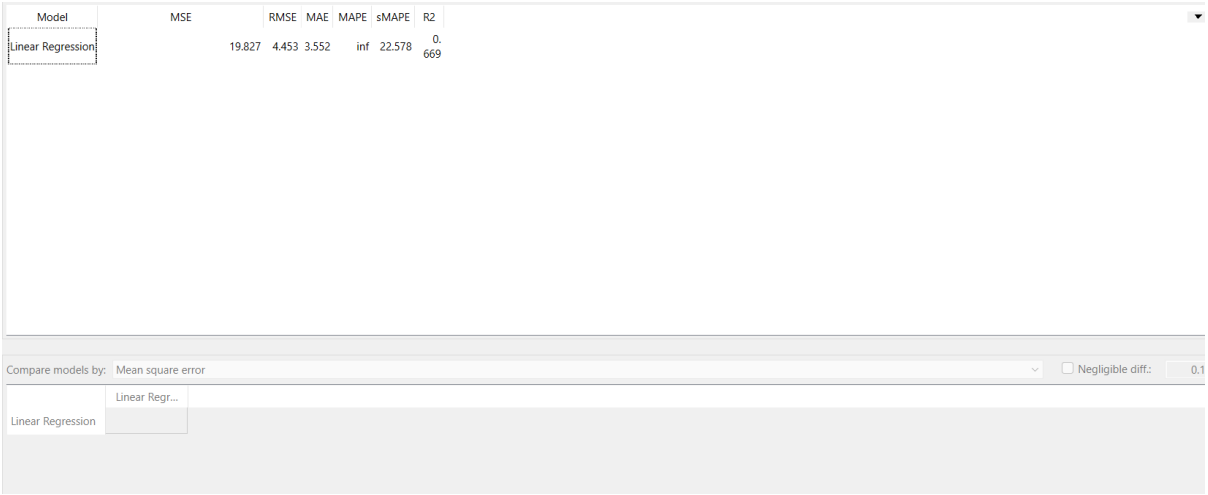
6: Performing Regression on Dataset

Aim: To perform regression on a dataset using Orange Data Mining by building a Linear Regression model and evaluating it with Test and Score.



Steps

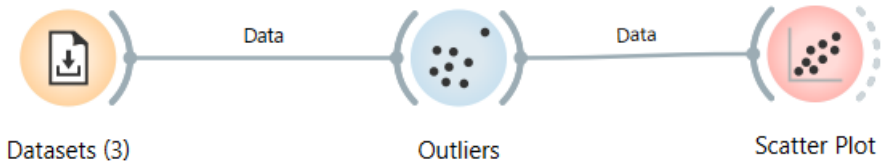
Output:



Result: Regression on Dataset was performed using Linear Regression in Orange (RMSE = 4.453, MAE = 3.552, R^2 = 0.669).

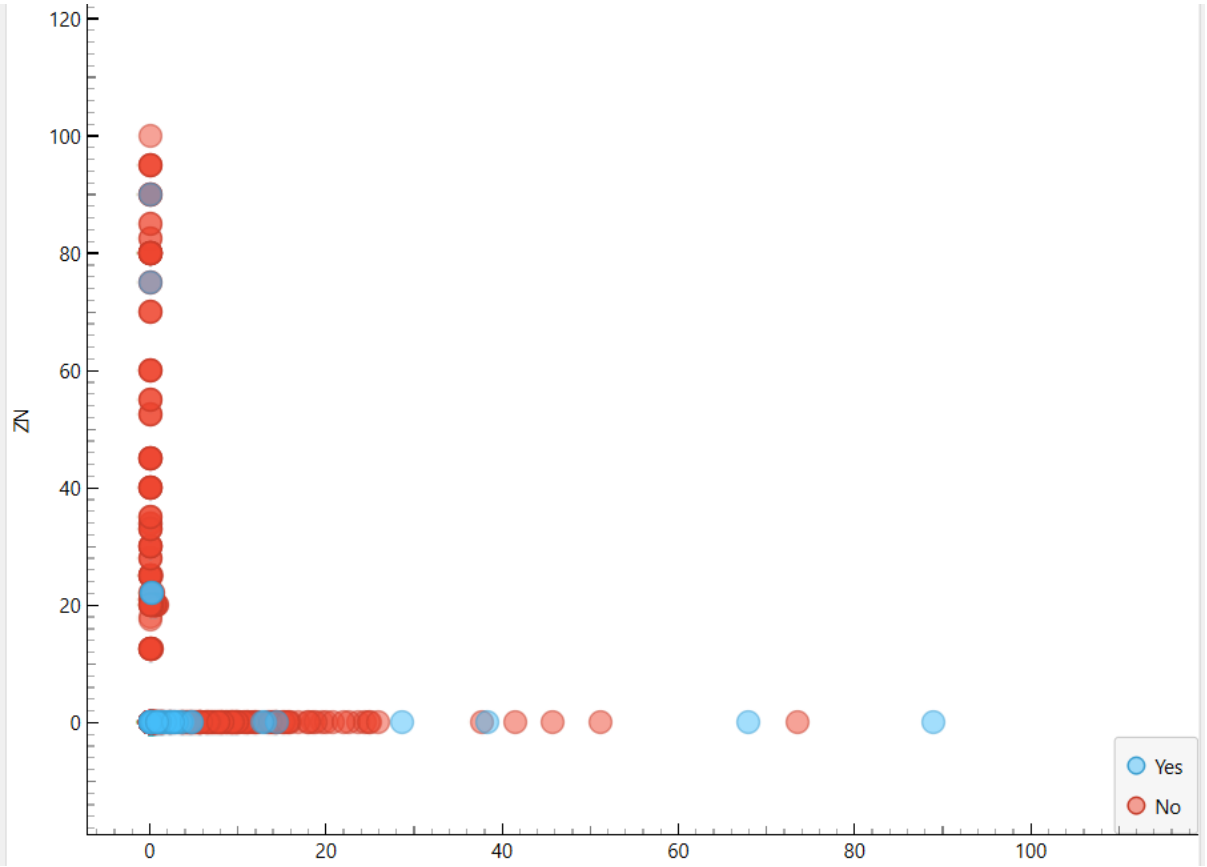
7: Implementation of Outlier Analysis using Orange

Aim: To detect outliers in a dataset using Orange Data Mining and visualize them with a scatter plot.



Steps

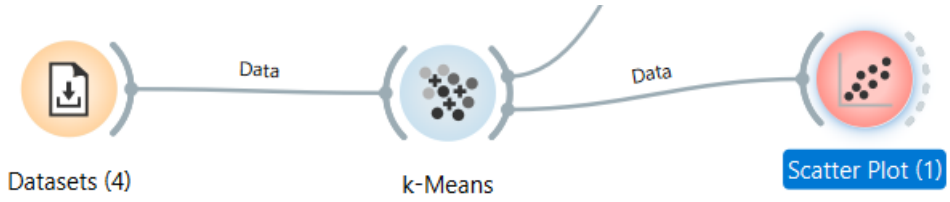
Output:



Result: Outlier Analysis using Orange was implemented.

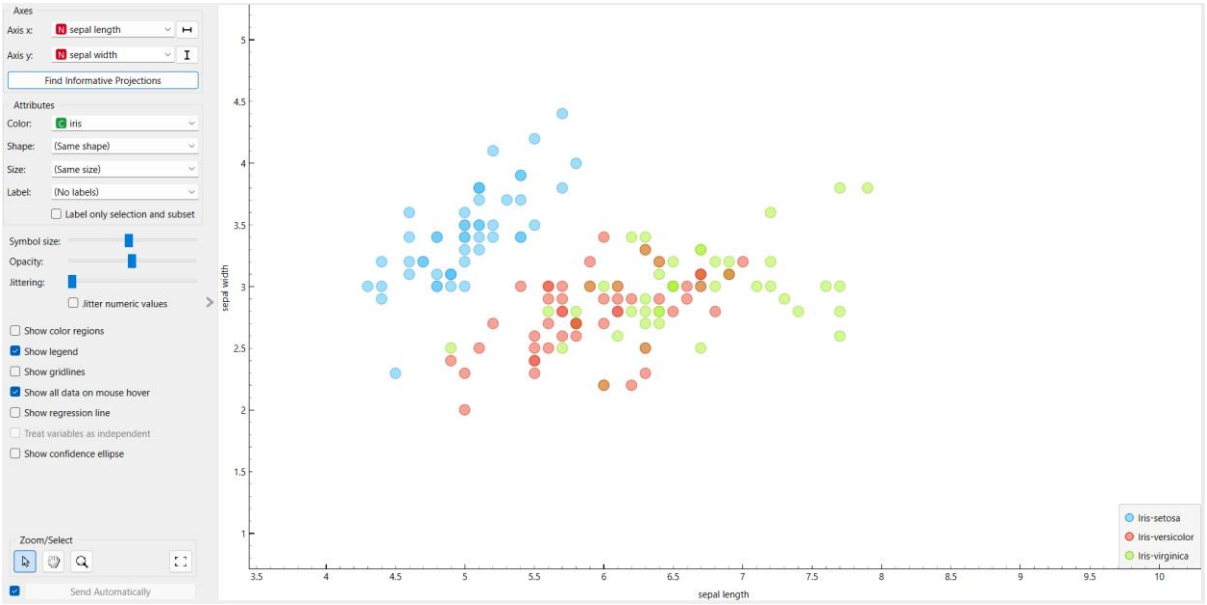
8: K-Means Algorithm for Iris Dataset

Aim: To apply the K-Means clustering algorithm on the Iris dataset using Orange Data Mining and visualize the result with a scatter plot.



Steps

Output:



Result: K-Means Algorithm for Iris Dataset was implemented using Orange.

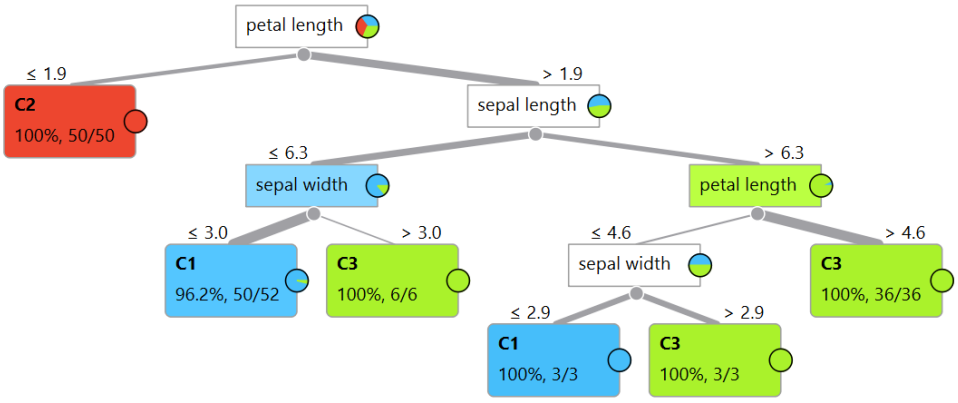
9: Clustering on iris.arff using Simple K-Means

Aim: To perform clustering on iris.arff using Simple K-Means in Orange Data Mining and interpret the resulting clusters (C1, C2, C3) using a classification tree.



Steps

Output:



Result: Clustering on iris.arff using Simple K-Means was achieved.